

Abstract No.

Tumor-driven seizures in pediatric low grade glioma: a retrospective study of the German cohort

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¹HIT-Loggic registry for pediatric low grade glioma

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Background:

Low-grade gliomas (LGG) represent the most common pediatric solid tumors. The HIT-LOGGIC registry accrues over 350 newly diagnosed pediatric LGG cases annually. Tumor-related epilepsy (TRE) is a common clinical symptom. In a subset of patients, seizures may be the first and occasionally, the only manifestation of the disease. Several factors have been proposed to influence the risk of TRE, including tumor location and specific neuropathological subtypes. This study aims to investigate potential clinical and histopathological risk factors associated with seizure occurrence.

Methods:

Clinical data are collected using standardized questionnaires and clinical reports. Integrated neuropathological diagnostics are performed at reference centers using epigenetic methylation analysis as well as DNA and RNA panel sequencing. Imaging evaluations are conducted at the registry's national reference center. The influence of potential clinical, radiological, and molecular factors on refractory epilepsy and on clinical outcome is analyzed.

Results:

We identified a cohort of 246 patients with pediatric LGG and TRE. The age at diagnosis ranged from 2 weeks to 19 years. In 85.3% of cases, epileptic seizures were the first clinical symptom. Glioneuronal tumors such as gangliogliomas and dysembryoplastic neuroepithelial tumors were the most common neuropathological entities, accounting for 65 (30.1%) and 47 (21.8%) cases, respectively. A BRAF V600E mutation was detected in 42% patients. 86.2% of tumors are located in the cerebral hemisphere, most frequently in the temporal lobe. 98 patients had treatment-resistant epilepsy. In 108 cases, surgical interventions were performed with the primary goal of achieving seizure freedom. In 76.9% of these cases, a Class 1 seizure outcome was achieved according to the classification of the International League Against Epilepsy.



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Conclusion:

Identifying potential risk factors for seizures is essential for early recognition and appropriate management. This work highlights the importance of interdisciplinary care in the treatment of pLGG, particularly in patients affected by TRE.